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VTU Question Papers

“Mechanical Engineering”

III, V & VII Semester

Dec/Jan 2024

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Bannur road, Alanahalli, Mysore-570028**

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Mechanics of Materials

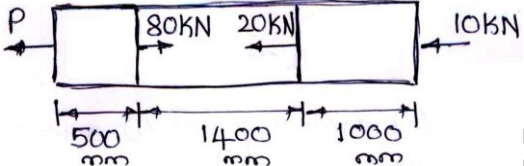
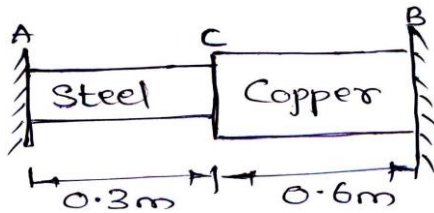
Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

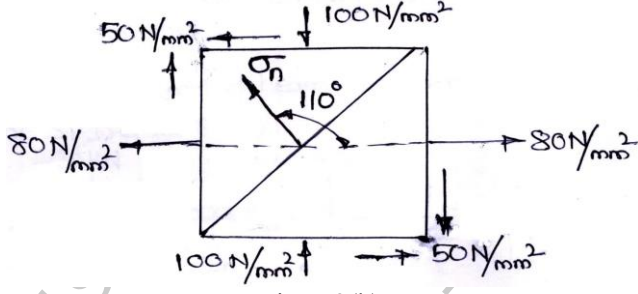
2. *VTU Formula Hand Book is permitted.*

3. *M*: Marks, *L*: Bloom's level, *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	State Hooke's law. Draw a neat diagram of stress-stain curve for mild steel and mark the salient points and zones.	5	L2	CO1
	b.	Derive an expression for elongation in a tapered bar of circular cross-section, subjected to an axial tensile load "F".	7	L3	CO1
	c.	A brass bar having uniform cross-section area of 300mm^2 is subjected to a load as shown in Fig.1(c). Find the total elongation of bar and the magnitude of load "P" if, $E = 84\text{GPa}$.  Fig.Q.1(c)	8	L3	CO1
OR					
Q.2	a.	Define the following: i) Poisons ratio ii) Bulk modulus iii) Factor of safety iv) True stress v) Hardness.	5	L2	CO1
	b.	A bar of 20mm diameter is tested in tension. It is observed that when a load of 37.7kN is applied, the extension measured over a gauge length of 200mm is 0.12mm and contraction in diameter is 0.0036mm. Find Poisson's ratio and elastic constants E, G, K.	7	L3	CO1
	c.	A stepped bar is fixed at its two ends rigidly. The bar is free from stresses when its temperature is 30°C. When the temperature is increased to 90°C, determine: i) Stresses induced in copper and steel portions. ii) Displacement at the junction point "C". Take $E_c = 100\text{GPa}$, $E_s = 200\text{GPa}$, $\alpha_c = 1.8 \times 10^{-5}/^\circ\text{C}$ and $\alpha_s = 1.2 \times 10^{-5}/^\circ\text{C}$, $A_s = 80\text{mm}^2$, $A_c = 120\text{mm}^2$.  Fig.Q.2(c)	8	L3	CO1

1 of 3

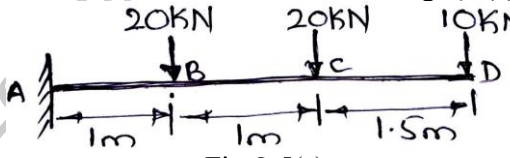
Module – 2

Q.3	a.	Define: i) Principal plane ii) Principal stress iii) Maximum shear stress iv) Plane of maximum shear.	8	L2	CO2
	b.	An element with the stresses acting on it is shown in Fig.Q.3(b), by Mohr's circle method find: i) Normal and shear stress acting on a plane whose normal is at an angle of 110° with respect to x-axis ii) Principal stresses and their locations. iii) Maximum shear stresses and their locations.  Fig.Q.3(b)	12	L3	CO2

OR

Q.4	a.	Derive expressions for circumferential and longitudinal strains in thin cylinder. Hence show that volumetric strain is $\epsilon_v = \frac{pd}{4tE} (5 - 4\gamma)$	8	L3	CO2
	b.	A cast iron pipe has 200mm internal diameter and 50mm metal thickness. It carries water at a pressure of 5 N/mm^2 . Calculate the intensities of circumferential and radial pressures. Sketch the stress distribution across the section.	12	L3	CO2

Module – 3

Q.5	a.	Discuss about different types of beams and loads.	6	L2	CO3
	b.	Obtain a relation between load intensity, shear force and bending moment.	6	L3	CO3
	c.	Draw the BMD and SFD for cantilever shown in Fig.Q.5(c).  Fig.Q.5(c)	8	L3	CO3

OR

Q.6	a.	Define: i) Point of contraflexure ii) Bending moment iii) Shear force.	6	L2	CO3
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	b.	A simply supported beam is shown in Fig.Q.6(b). Draw the SFD and BMD.	14	L3	CO3
Fig.Q.6(b)					
Module – 4					
Q.7	a.	List the assumptions made in theory of pure bending. Derive the bending equation with usual notations.	10	L3	CO4
	b.	A simply supported beam of 5m span has a cross-section 150mm × 250mm. If the permissible stress is 10N/mm ² . Find: i) Maximum UDL intensity ii) Maximum concentrated load “P” at 2m from one end.	10	L3	CO4
OR					
Q.8	a.	A uniform I-section beam is subjected 100kNm bending moment. Plot the stress variation across the section.	10	L3	CO4
Fig.Q.8(a)					
	b.	A cantilever of square section 200mm × 200mm and length 2m, fails in flexure when 12kN is placed at free end. A rectangular beam of same material and simply supported over length of 3m, 150mm wide and depth 300mm. Calculate minimum central concentrated load required to break the beam.	10	L3	CO4
Module – 5					
Q.9	a.	Derive torsion equation. Also list the assumptions.	10	L3	CO5
	b.	Find the shaft diameter required to transmit 60kW at 150rpm, if maximum torque is 25% more than mean torque for a maximum shear stress of 60MPa. Find angle of twist for a 4m length. Take, G = 80GPa.	10	L3	CO5
OR					
Q.10	a.	Derive an expression for critical load in a column subjected to compressive load, when one end fixed and other free.	10	L3	CO5
	b.	A 1.5m long column has a circular cross-section of 50mm diameter. One end of column is fixed and other end is free. Taking FOS = 3. Calculate safe load using i) Rankines formula, yield stress 560N/mm ² and $a = \frac{1}{1600}$ ii) Eulers formula, E = 120GPa.	10	L3	CO5

Third Semester B.E./B.Tech Degree Examination, Dec.2023/Jan.2024 Manufacturing Process

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define a gating system. Explain with sketches types of gating system.	10	L1	CO1
	b.	Explain with a neat sketch CO ₂ moulding process.	10	L2	CO1
OR					
Q.2	a.	Define pattern and explain with a neat sketches any four pattern allowances.	10	L1	CO1
	b.	Explain in detail the procedure to determine grain fineness number of greens and in foundry lab.	10	L2	CO1
Module – 2					
Q.3	a.	Explain with a neat sketch coreless induction furnace.	10	L2	CO2
	b.	Explain with a neat sketch cupola furnace.	10	L2	CO2
OR					
Q.4	a.	Explain with a neat sketch centrifugal casting process.	10	L2	CO2
	b.	Explain with neat sketches casting defects.	10	L2	CO2
Module – 3					
Q.5	a.	Illustrate the following metal forming processes with neat sketches : i) Bending ii) Piercing iii) Blanking.	10	L2	CO3
	b.	Explain the following yield criteria : i) Tresca yield criteria ii) Von-Mises yield criteria.	10	L2	CO3
OR					
Q.6	a.	Describe compound and progressive die processes.	10	L2	CO3
	b.	Explain the importance of temperature in metal forming and write the differences between hot working and cold working.	10	L2,1	CO3
Module – 4					
Q.7	a.	Explain with neat sketches types of flames produced in OXY-Acetylene welding.	10	L2	CO4
	b.	Explain with a neat sketch OXY-Acetylene gas welding process.	10	L2	CO4
OR					
Q.8	a.	Explain with a neat sketch MIG welding and mention its advantages, disadvantages and applications.	10	L2	CO4
	b.	Explain with a neat sketch Manual metal arc welding and also mention its advantages, disadvantages and applications.	10	L2	CO4
Module – 5					
Q.9	a.	Explain with neat sketches welding defects.	10	L2	CO5
	b.	Explain with a neat sketch residual stresses in welded structures.	10	L2	CO5
OR					
Q.10	a.	Describe the following : i) Soldering ii) Brazing.	10	L2	CO5
	b.	Explain with a neat sketch resistance welding process.	10	L2	CO5

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Third Semester B.E./B.Tech Degree Examination, Dec.2023/Jan.2024 Material Science and Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Calculate the APF for FCC and BCC unit cell in crystal structure.	10	L3	CO1
	b.	Enumerate the type of crystal imperfections and explain briefly with a suitable sketch grain boundary and twin boundary defects.	10	L2	CO1
OR					
Q.2	a.	Explain briefly with suitable sketches the plastic deformation by Slip and Twinning.	10	L2	CO1
	b.	Define the following terms : i) Unit cell and space lattice ii) Coordination number.	5	L2	CO1
	c.	Molybdenum has BCC and a density of $10.2 \times 10^3 \text{ kg/m}^3$. Calculate its atomic radius. The atomic weight of molybdenum is 95.94gm/mol. $N_A = 6.023 \times 10^{23} \text{ atoms/mol}$.	5	L3	CO1
Module – 2					
Q.3	a.	Draw neatly the solid solution binary phase diagram of a Ni-Cu system and explain briefly.	10	L2	CO2
	b.	State and explain briefly the Fick's 1 st and 2 nd law of diffusion.	10	L2	CO2
OR					
Q.4	a.	Explain briefly with a neat sketch the eutectic system of two components completely soluble in liquid state and partially soluble in solid state.	08	L2	CO2
	b.	Draw a neat sketch of iron-carbon equilibrium diagram and show all phases on the diagram also show the three invariant reactions.	12	L2	CO2
Module – 3					
Q.5	a.	Explain briefly mechanism of solidification with suitable sketches.	10	L2	CO3
	b.	With a suitable sketch explain normalizing heat treatment process.	10	L2	CO3
OR					
Q.6	a.	Draw a neat labeled Time-Temperature Transformation [TTT] diagram for Eutectoid steel (0.8%C) and explain briefly.	10	L3	CO3
	b.	With a neat sketch briefly explain Austempering and Martempering heat treatment process.	10	L2	CO3
Module – 4					
Q.7	a.	With a flow chart, explain briefly the powder metallurgy process and its applications.	10	L2	CO4
	b.	Enumerate the different powder production methods, with suitable sketch explain atomization method.	10	L2	CO4
OR					
Q.8	a.	Explain briefly thermal spray coating with suitable sketch. Mention the advantages of surface coatings and treatments.	10	L2	CO4
	b.	Advantages and limitations of powder metallurgy process.	10	L2	CO4

Module – 5					
Q.9	a.	Classify engineering metals. Enumerate the types of cast iron and mention the compositions, properties and applications.	10	L2	CO5
	b.	With a neat sketch, explain the production of composite by : i) Filament winding process ii) Bag molding process.	10	L2	CO5
OR					
Q.10	a.	With a suitable sketch explain the production of metal matrix composite by stir casting process.	10	L2	CO5
	b.	What are the factors affecting the selection of materials explain briefly.	10	L2	CO5

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Basic Thermodynamics

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Use of thermodynamic data handbook and steam tables is permitted.
4. Assume missing data suitably.

Module – 1			M	L	C
Q.1	a.	State Zeroth law of thermodynamics and justify how it forms the basis for temperature measurement.	05	L1	CO1
	b.	Derive an expression for P-dV work for a process in which (i) $PV = C$ (ii) $PV^n = C$ where C is a constant.	05	L2	CO1
	c.	The temperature scale by a certain thermometer is given by the relation $t = a \ln x + b$ where 'a' and 'b' are constants and x is the thermometric property of the fluid in the thermometer. If at ice and steam points the thermometric property is found to be 1.5 and 7.5 respectively. What will be the temperature corresponding to the thermometric property 3.5?	10	L3	CO1
OR					
Q.2	a.	Show that thermodynamics definition for work is superior to mechanics definition.	05	L1	CO1
	b.	With a neat sketch, explain working principle of constant volume gas thermometer.	05	L2	CO1
	c.	A perfect gas is undergoing a process in which $T \propto V^{2/5}$. Calculate the work done by the gas in going from state 1 in which the pressure is 100 bar and volume is 4 m ³ to the state 2 in which volume is 2 m ³ . Also calculate the final pressure.	10	L4	CO1
Module – 2					
Q.3	a.	State the first law of thermodynamics along with the mathematical expression for the following: (i) A closed system undergoing a cycle (ii) A closed system undergoing a change of state.	05	L1	CO2
	b.	With a neat sketch of steady flow device, write the steady flow energy equation with usual notations.	05	L1	CO2
	c.	A stationary mass of gas is compressed without friction from an initial state of 0.3 m ³ and 0.105 MPa to a final state of 0.15 m ³ and 0.105 MPa, the pressure remaining constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas change?	10	L3	CO2
OR					
Q.4	a.	Write the steady flow energy equation for (i) Boiler (ii) Centrifugal pump.	05	L1	CO2
	b.	Show that energy is a property of the system.	05	L2	CO2
	c.	In a certain steady flow process, 12 kg of fluid per minute enters at a pressure of 1.4 bar, density 25 kg/m ³ , velocity 120 m/s and internal energy 920 kJ/kg. The fluid properties at exit are 5.6 bar, density 5 kg/m ³ , velocity 180 m/s, and internal energy 720 kJ/kg. During the process, the fluid rejects 60 kJ/s of heat and rises through 60 m. Determine work done during the process in KW.	10	L4	CO2

Module – 3					
Q.5	a.	Define thermal efficiency of a heat engine and COP of a refrigerator along with mathematical expressions for both. Write their schematic diagram.	05	L1	CO3
	b.	Define entropy and show that it is a property of the system.	05	L2	CO3
	c.	A reversible heat engine converts one-sixth of the heat input into work. When the temperature of the sink is reduced by 62°C, its efficiency is doubled. Find the temperature of the source and the sink.	10	L4	CO3
OR					
Q.6	a.	State and prove Clausius theorem.	05	L2	CO3
	b.	Give the Kelvin-Planck and Clausius statements of the second law of thermodynamics.	05	L1	CO3
	c.	A lump of steel of mass 10 kg at 627°C is dropped in 100 kg of oil at 30°C. The specific heats of steel and oil are 0.5 kJ/kgK and 3.5 kJ/kgK respectively. Calculate the entropy change of steel, the oil and the universe.	10	L3	CO3
Module – 4					
Q.7	a.	Define Available Energy (AE) and Unavailable Energy (UE). Show that unavailable energy is the product of lowest temperature of heat rejection and the change in entropy of the system during the process of supplying heat. Draw the necessary schematics and T-S diagrams.	10	L4	CO4
	b.	With a neat sketch, explain the working principle of separating and throttling calorimeter.	10	L2	CO4
OR					
Q.8	a.	In a certain process, a vapor, while condensing at 420°C, transfers heat to water evaporating at 250°C. The resulting steam is used in a power cycle which rejects heat at 35°C. What is the fraction of the available energy in the heat transfer process from the vapour at 420°C that is lost due to the irreversible heat transfer at 250°C?	10	L3	CO4
	b.	Draw the phase equilibrium diagram for a pure substance on P-T coordinates and show the fusion curve, vaporization curve, sublimation curve, triple point and critical point.	10	L2	CO4
Module – 5					
Q.9	a.	Write notes on: (i) Daltons law of partial pressure (ii) Amagots law of additive volumes (iii) Compressibility factor (iv) Law of corresponding states (v) Generalized compressibility chart	10	L2	CO5
	b.	One kg of ideal gas is heated from 50°C to 150°C. Determine: (i) Change in internal energy (ii) Change in enthalpy (iii) Change in flow energy (iv) $\bar{C}_v$ and $\bar{C}_p$ Take $R = 280 \text{ kJ/kgK}$; $\gamma = 1.32$ for gas.	10	L3	CO5
OR					
Q.10	a.	Derive: (i) Maxwell's equations (ii) The first and second Tds equations.	10	L2	CO5
	b.	Find the gas constant and apparent molar mass of a mixture of 2 kg O ₂ and 3 kg N ₂ given that the universal gas constant is 8314.3 J/kgK, molar mass of O ₂ and N ₂ are respectively 32 and 28.	10	L3	CO5

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Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024

Theory of Machines

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms :

(i) Kinematic pair.	(ii) Kinematic chain.
(iii) Mechanism	(iv) Degree of freedom.
(iv) Inversion.	

(05 Marks)
- b. With neat sketch, explain any three inversions of four-bar kinematic chain. (09 Marks)
- c. A four bar chain mechanism ABCD is made up of four links, pin jointed at the ends. AD is fixed line which is 120 mm long. The links AB, BC and CD are 60 mm, 80 mm and 80 mm respectively. At a certain instant, the link AB makes an angle of 60° with the link AD. If the link AB rotates at a uniform speed of 10 rpm clockwise, determine angular velocity of the link BC and CD. (06 Marks)

OR

- 2 a. Distinguish between machine and structure. (Any four) (04 Marks)
- b. The Fig. Q2 (b) shows a toggle mechanism, Crank OP rotates at a uniform speed of 120 rpm in clockwise direction. Determine the velocity and acceleration of the slider S. The length of the various links are OP = 80 mm, PR = 180 mm, QR = 240 mm and SR = 270 mm. (08 Marks)

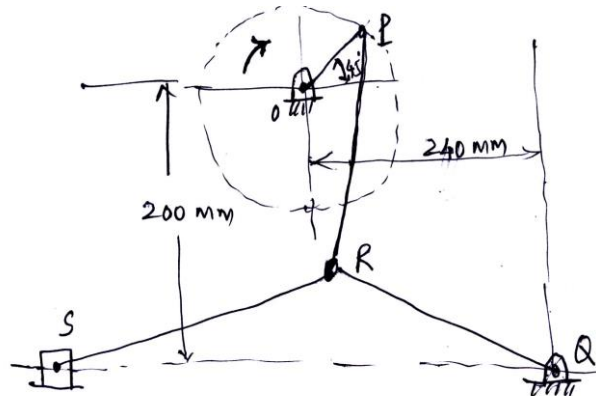


Fig. Q2 (b)

- c. Using complex algebra, determine the velocity and acceleration of the piston, angular acceleration of the connecting rod of a reciprocating engine. If the crank length is 50 mm, connecting rod 200 mm, crank speed is constant at 3000 rpm and crank angle is 30° . (08 Marks)

Module-2

- 3 a. Explain the static equilibrium of two forces, three forces and member with two forces and a torque. (06 Marks)
- b. State D'Alemberts principle. When a crank 45° from inner dead centre on the down stroke. The effective steam pressure on the piston of a vertical steam engine is 2.5 bar. The diameter of the cylinder = 0.75 m, stroke of the piston = 0.50 m and the length of the connecting rod = 1 m. Determine the torque on the crank shaft, if the engine runs at 350 rpm and the mass of reciprocating parts is 200 kg. (06 Marks)

- c. What is fly wheel? The turning moment diagram for a multicylinder engine has been drawn to a scale of 1 mm = 600 N-m vertically and 1 mm = 3° horizontally. The intercepted areas between the output torque curve and the mean resistance line taken in order from one end are as follows : +52, -124, +92, -140, +85, -72 and +107 mm².

When the engine is running at a speed of 600 rpm, if the total fluctuation of speed is not to exceed +1.5% of the mean. Find the necessary mass of the flywheel of radius 0.5 m.

(08 Marks)

OR

- 4 a. For the mechanism shown in Fig. Q4 (a), find the required input torque for static equilibrium. The lengths OA and AB are 250 mm and 650 mm respectively. $F = 500$ N.

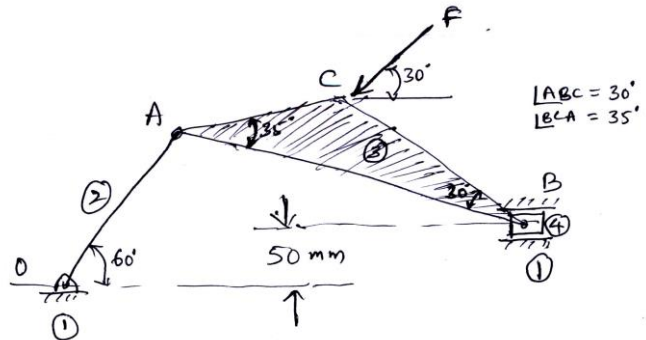


Fig. Q4 (a)

(10 Marks)

- b. The crank pin circle radius of a horizontal engine is 300 mm. The mass of reciprocating parts is 250 kg. when the crank has travelled 60° for IDC. The difference between the driving and the back pressure is 0.35 N/mm². The connecting rod length between centres is 1.2 m and the cylinder bore is 0.5 m. If the engine runs at 250 rpm and if the effect of piston rod diameter is neglected, calculate : (i) Pressure on the slide bars (ii) Thrust in the connecting rod (iii) Tangential force on the crank pin and (iv) Turning moment on the crank shaft.

(10 Marks)

Module-3

- 5 a. State law of gearing. Derive an expression for the minimum number of teeth on the pinion in order to avoid interference in involute gear teeth when it meshed with wheel. (08 Marks)
- b. Two mating gears with module pitch 6 mm have 20 and 50 teeth of pressure angle 30° and addendum 6 mm. Determine the number of pairs of teeth in contact. (05 Marks)
- c. What is a Gear Train? With neat sketch, explain different types of gear trains. (07 Marks)

OR

- 6 a. With neat sketch, explain spur gear terminology. (06 Marks)
- b. Two 20° involute spur gears mesh externally to give a velocity ratio of 3. Module is 3 mm and the addendum is equal to 1.1 times the module. If the pinion rotates at 120 rpm ; determine ; Minimum number of teeth on each wheel to avoid interference. (04 Marks)
- c. An epicyclic gear train consists of a sun wheel (S) a stationary internal gear (E) and the three identical planet wheel (P) carried on a star shaped planet carrier (C). The sizes of different toothed wheels are such that the planet carrier C rotates at $\frac{1}{5}$ of the speed of the sun wheel.

The minimum number of teeth on any wheel is 16. The driving torque on the sun wheel is 100 N-m. Determine :

- (i) Number of teeth on different wheels of train.
- (ii) Torque necessary to keep the internal gear stationary.

(10 Marks)

Module-4

- 7 a. Three masses of 8 kg, 12 kg and 15 kg attached at radial distance of 80 mm, 100 mm and 60 mm respectively to a disc on a shaft are in static balance. Determine the angular positions of masses 12 kg and 15 kg relative to 8 kg mass. (06 Marks)
- b. Explain why only part of unbalanced forces due to reciprocating masses is balanced by revolving masses. (06 Marks)
- c. What is a governor? Derive an expression for the equilibrium speed of a porter governor. (08 Marks)

OR

- 8 a. A rotating shaft carries four masses 1, 2, 3 and 4 which are radially attached to it. The mass centers are 30 mm, 38 mm, 40 mm and 35 mm respectively from the axis of rotation. The masses 1, 3 and 4 are 7.5, 5 and 4 kg respectively. The axial distance between the planes 1 and 2 is 400 mm and between 2 and 3 is 500 mm. The masses 1 and 3 are at right angles to each other. Find for complete balance
- Angle between 1, 2 and 1, 4
 - Axial distance between 3 and 4.
 - Magnitude of mass 2.
- (10 Marks)
- b. The radius of rotation of the balls of a Hartnell Governor is 8 cm at the minimum speed of 300 rpm. Neglecting gravity effect determine the speed after the sleeve is lifted by 6 cm ; also determine the initial compression of the spring, governor effort and power. The particulars of the governor are, length of ball arm = 15 cm, Length of sleeve arm = 10 cm, Mass of each ball = 4 kg and stiffness = 25000 N/m. (10 Marks)

Module-5

- 9 a. Define the terms : (i) Damping (ii) Damping ratio (iii) Stiffness of the spring (iv) Logarithmic decrement. (06 Marks)
- b. Determine the natural frequency of the simple pendulum by using Newton's method. Neglecting the mass of rod. (04 Marks)
- c. Determine the critical speed when an automobile trailer is travelling over a road with sinusoidal profile of wavelength 15 meter as shown in Fig. Q9 (c) and amplitude of 75 mm. The spring of the automobile are compressed 0.125 m under its own weight. Also determine the amplitude of vibration at 50 kmph.

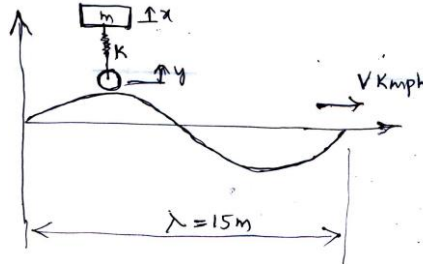


Fig. Q9 (c)

(10 Marks)

OR

- 10 a. Derive an expression for steady state solution with viscous damping due to harmonic force. (10 Marks)
- b. Determine the damped natural frequency for the system shown in Fig. Q10 (b). Also if $m = 1.5$ kg, $K = 4900$ N/m, $a = 10$ cm and $b = 13$ cm, determine the critical damping co-efficient 'C' for the system. (10 Marks)

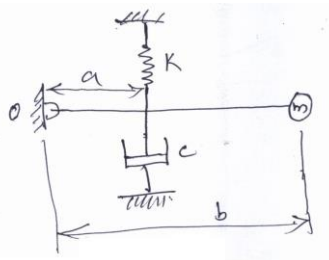


Fig. Q10 (b)

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21ME52

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024

Thermo-Fluids Engineering

Time: 3 hrs.

Max. Marks: 100

Note:1. Answer any FIVE full questions, choosing ONE full question from each module.

2. Use of thermodynamic data hand book is permitted.

Module-1

- 1
 - a. With suitable formulae, explain Morse test for measuring frictional power in an IC engine. (05 Marks)
 - b. What are the parameters to be considered while framing heat balance sheet for an IC engine? (05 Marks)
 - c. The following data refers to a four stroke diesel engine :
 Cylinder diameter = 200 mm,
 Stroke = 300 mm, Speed = 300 rpm,
 Effective brake load = 500 kg,
 Mean circumference of the brake drum = 400 mm,
 Mean effective pressure = 6 bar,
 Diesel consumption = 0.1 lt/min,
 Calorific value = 43900 kJ/kg,
 Specific gravity = 0.78.
 Find :
 (i) Brake power
 (ii) Indicated power
 (iii) Frictional power
 (iv) Mechanical efficiency
 (v) Indicated thermal efficiency. (10 Marks)

OR

- 2
 - a. Explain the procedure of conducting experiment to evaluate the performance of a reciprocating 2 stage air compressor. (06 Marks)
 - b. With the help of PV diagram, derive an expression for the volumetric efficiency in terms of clearance ratio for a 2 stage reciprocating air compressor. (06 Marks)
 - c. A double acting air compressor of 18 cm diameter and 120 cm stroke runs at 120 rpm and operates between 1 bar and 10 bar, the lower temperature being 15°C. Estimate the power, final temperature and temperature rise, if the compression index = 1.3. (08 Marks)

Module-2

- 3
 - a. With a neat sketch, explain vapor absorption refrigerator. (06 Marks)
 - b. Mention a few properties of a good refrigerant. (04 Marks)
 - c. An air refrigerator was designed to produce 80 tons of refrigeration, with air entering the compressor at 8°C. The air was cooled after compression in a cooler to 27°C. It was observed that the actual power required was 20% more than theoretical power with an air circulation rate of 2 kg/s. Determine (i) The theoretical COP (ii) Actual COP (iii) Power required to run the compressor. Assume $r = 1.4$, $C_p = 1.005$ kJ/kgK for air and the cycle is ideal. (10 Marks)

OR

- 4 a. Define the terms :
- Dew point temperature.
 - Dry bulb temperature.
 - Wet bulb temperature.
 - Humidity ratio.
 - Relative humidity.
- (10 Marks)
- b. An auditorium of 150 seating capacity is conditioned for the following specifications:
 Outdoor conditions = 40°C DBT and 20°C WBT,
 Required indoor conditions = 20°C DBT and 60% RH.
 Amount of outdoor air supplied = $0.4\text{ m}^3/\text{min}$ per person.
 If the required condition is achieved first by adiabatic humidification and then by cooling.
 Calculate : (i) Capacity of the cooling coil in tones
 (ii) The capacity of the humidifier in kg/min.
- (10 Marks)

Module-3

- 5 a. With a neat sketch, explain various parts of turbomachine. (06 Marks)
- b. With the proper velocity triangles derive an expression for alternate form of Euler's turbine equation. (06 Marks)
- c. The velocity of steam in a De-Laval turbine at the inlet is 1200 m/s . The nozzle angle at the inlet is 22° and rotor blades are equiangular. Assuming relative velocities of the fluid at the inlet and exit to be equal and tangential speed of the rotor is 400 m/s . Determine
- The blade angles at the inlet and exit.
 - Power developed if mass flow rate is 1 kg/s
 - Tangential force exerted on the blade ring
 - Utilization factor.
- (08 Marks)

OR

- 6 a. Compare turbomachines and positive displacement machines. (06 Marks)
- b. With a neat sketch, explain the working of a gear pump. (06 Marks)
- c. In a radial inward flow turbine, the runner outer diameter is 75 cm and the inner diameter is 50 cm . The runner speed is 400 rpm . Water enters the runner at a velocity of 15 m/s at an angle of 15° to wheel tangent at inlet. The flow is radial at exit with a velocity of 5 m/s . Find the blade angles at inlet and exit. Also determine the power output for flow rate of $1.5\text{ m}^3/\text{s}$, degree of reaction and utilization factor. (08 Marks)

Module-4

- 7 a. Explain the experimental procedure to evaluate the efficiency of a Pelton wheel. (06 Marks)
- b. Derive an expression for hydraulic efficiency of a Pelton turbine. (06 Marks)
- c. The external and internal diameters of inward flow reaction turbine are 1.2 m and 0.6 m respectively. The head on turbine is 22 m and velocity of flow through the turbine is constant and is equal to 2.5 m/s . The guide blade angle is 10° and runner vanes are radial at inlet. If the discharge at outlet are radial. Determine
- Speed of turbine
 - Vane angle at outlet of runner.
 - Velocity triangles at inlet and exit
 - Hydraulic efficiency.
- (08 Marks)

OR

- 8 a. Derive expression for theoretical head capacity relationship of a centrifugal pump for different vane angles. (08 Marks)
- b. With a sketch, show different parts of a centrifugal pump. (04 Marks)
- c. A centrifugal pump is designed to run at 1450 rpm. With minimum discharge of 1800 litres/min against a total head of 20 m. The suction and delivery pipes are designed such that they are equal in size of 100 mm. If the inner and outer diameters of the impeller are 12 cm and 24 cm respectively. Determine the blade angles β_1 and β_2 for radial entry neglecting friction and other losses. (08 Marks)

Module-5

- 9 a. Briefly explain the terms in relation with a centrifugal compressor,
 (i) Diffuser
 (ii) Slip factor. (06 Marks)
- b. Derive an expression for pressure ratio of a centrifugal compressor. (06 Marks)
- c. A centrifugal compressor delivers 30 kg/s of air with a total head pressure ratio of 4 : 1. The speed of the compressor is 12000 rpm, inlet total temperature is 15°C , stagnation pressure at inlet 1 bar, slip factor is 0.9, power input factor is 1.04, efficiency 80%. Calculate the outer diameter of the impeller. (08 Marks)

OR

- 10 a. Derive an expression for maximum blade efficiency for impulse turbine. (10 Marks)
- b. A simple impulse turbine has a mean blade speed of 200 m/s. The nozzles are inclined at 20° to the plane of rotation of the blades. The steam velocity from nozzles is 600 m/s. The turbine uses 3500 kg/hr of steam. The absolute velocity at exit is along the axis of the turbine. Determine, (i) Inlet and exit angles of blades (ii) Power output of turbine (iii) Diagram efficiency. (10 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024

Finite Elements Analysis

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain briefly the general steps involved in finite element method. (10 Marks)
- b. Explain convergence criteria and discretization process in F.E.M. (10 Marks)

OR

- 2 a. Explain 1D, 2D and 3D elements in F.E.M. (10 Marks)
- b. Explain plane stress and plane strain conditions. (10 Marks)

Module-2

- 3 a. Derive shape functions for 1D Quadratic bar elements in natural coordinates. (10 Marks)
- b. Derive shape functions for constant strain triangle, in natural coordinates. (10 Marks)

OR

- 4 A stepped bar as shown in Fig Q4. Determine the nodal displacement and stresses at each node. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

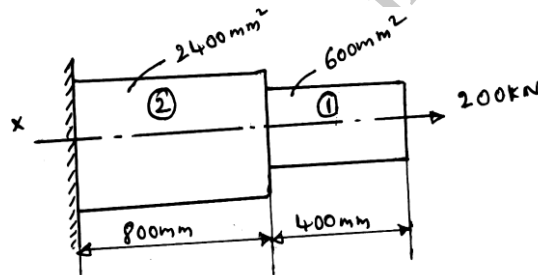


Fig Q4

(20 Marks)

Module-3

- 5 a. Derive H_1 and H_2 Hermite shape functions for beam elements. (10 Marks)
- b. A cantilever beam subjected to point load of 250 kN as shown in Fig Q5(b). Determine the deflection at free end. Take $E = 200 \text{ GPa}$, $I = 4 \times 10^6 \text{ mm}^4$.

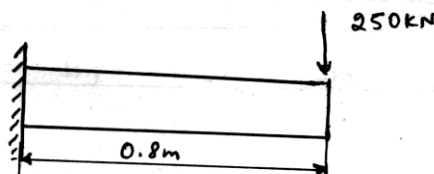


Fig Q5(b)

(10 Marks)

OR

- 6 a. Derive stiffness matrix equation for torsion of shaft. (10 Marks)
- b. A solid stepped bar of circular cross section as shown in Fig Q6(b), is subjected to a torque of 1 kN-m at its free end and a torque of 3 kN-m at its change in C/S. Determine the angle of twist in the bar. Take $E = 2 \times 10^5 \text{ MPa}$, $G = 7 \times 10^4 \text{ MPa}$.

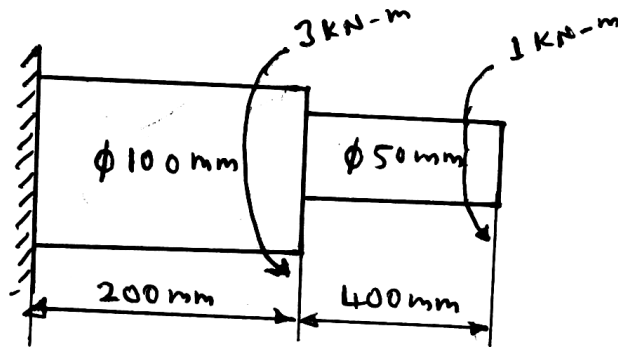


Fig Q6(b)

(10 Marks)

Module-4

- 7 Determine the temperature distribution in the wall using 1D heat elements. Give $K_1 = 25 \text{ W/m}^\circ\text{C}$, $K_2 = 35 \text{ W/m}^\circ\text{C}$, $h = 30 \text{ W/m}^2\text{C}$, $T_\infty = 900^\circ\text{C}$, $A = \text{unit area}$.

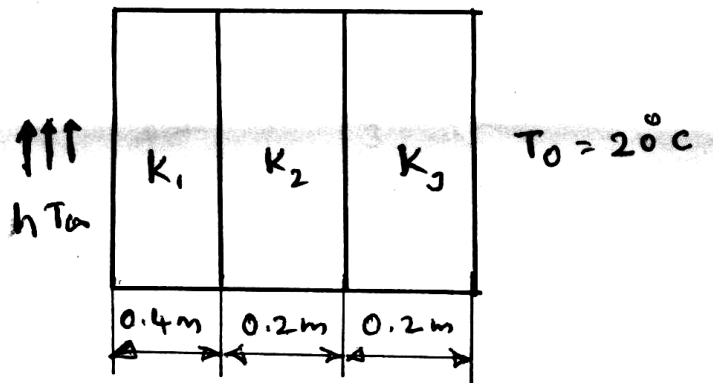


Fig Q7

(20 Marks)

OR

- 8 For smooth pipe of variable c/s shown in Fig Q8. Determine potential at junctions, Velocities in each section of pipe and volumetric flow rate. Potential at left end $P_1 = 10 \text{ m}^2/\text{s}$, right end $P_4 = 1 \text{ m}^2/\text{sec}$, fluid flow through pipe $K_x = 1$.

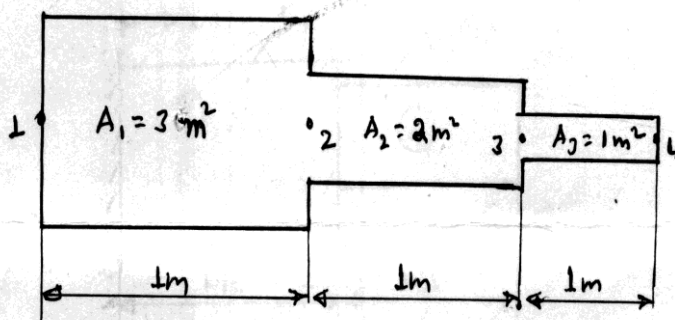


Fig Q8

(20 Marks)

Module-5

- 9 a. Derive stiffness, matrix of axisymmetric nodes with triangular elements. (10 Marks)
b. Evaluate nodal forces used to replace the linearly varying surface traction shown in Fig Q9(b)

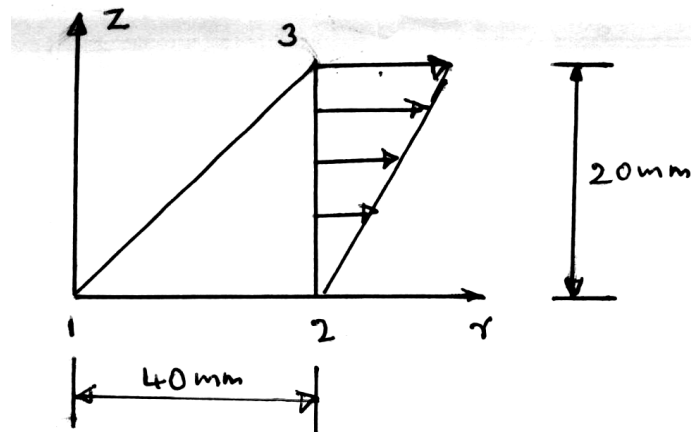


Fig Q9(b)

(10 Marks)

OR

- 10 Find Eigen values and Eigen vectors for stepped bar when it is subjected to axial vibration with fixed free end condition as shown in Fig Q10.

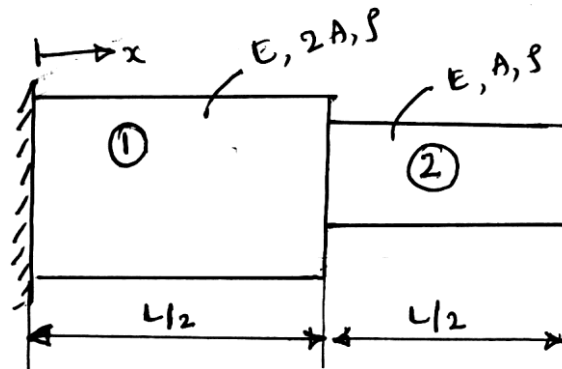


Fig Q10

(20 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024
Modern Mobility and Automotive Mechanics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List the components of Automotive Engine. Mention their function and materials used for manufacturing. (10 Marks)
- b. With neat sketch, explain the working of thermosymptom cooling system. (10 Marks)

OR

- 2 a. Explain dry sump lubrication system, with help of neat sketch. (10 Marks)
- b. Sketch and explain Electronic [Battery] Ignition system. (10 Marks)

Module-2

- 3 a. With the help of neat sketch, explain Multi – Plate Clutch. (10 Marks)
- b. With neat sketch, explain Torque converter. (10 Marks)

OR

- 4 a. Explain with neat sketch about the working of synchromesh gear box. (10 Marks)
- b. With a neat sketch, explain the working principle of Air suspension system. (10 Marks)

Module-3

- 5 a. What is Steering geometry? Define the following : i) Camber ii) Caster
 iii) King – pin inclination iv) Toe – in Toe - out. (10 Marks)
- b. What is ABS? Explain with appropriate sketch. (10 Marks)

OR

- 6 a. Explain the working of Power steering. (10 Marks)
- b. Explain about the safety terms : i) Air bags ii) Seat belt iii) Defogger and spoiler
 iv) Fire safety measures. (10 Marks)

Module-4

- 7 a. What is Pollution? Explain the exhaust gas pollutants effects on Environment. (10 Marks)
- b. Explain about the IC engine fuels and its advantages and disadvantages. (10 Marks)

OR

- 8 a. What is CNG vehicles? Explain its operation , advantages and disadvantages. (10 Marks)
- b. With the layout of electric hybrid vehicles, explain its operation and function of transmission and control system. (10 Marks)

Module-5

- 9 a. Explain the principle of Electric vehicles of 4 wheels. (10 Marks)
- b. What is Motor? Explain different types of motor and construction and working of any one. (10 Marks)

OR

- 10 a. Explain the construction and working of : i) lead batteries ii) sodium based batteries. (10 Marks)
- b. What re requirements of battery charging? How fire safety measuring in EV vehicles. (10 Marks)

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18ME71

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Control Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define control system. With example, explain open loop control system and closed loop control system. (10 Marks)
- b. List the types of controllers and explain PID controller with block diagram. (10 Marks)

OR

- 2 a. A thermometer is dipped in a vessel containing a liquid at constant temperature " $\theta_i(t)$ " with thermal capacitance " C " and Thermal Resistance (R). Temperature indicated by thermometer is " $\theta_0(t)$ ". Develop a transfer function for the system. (10 Marks)
- b. Develop a transfer function $\frac{X_2(s)}{F(s)}$ for the Fig.Q2(b).

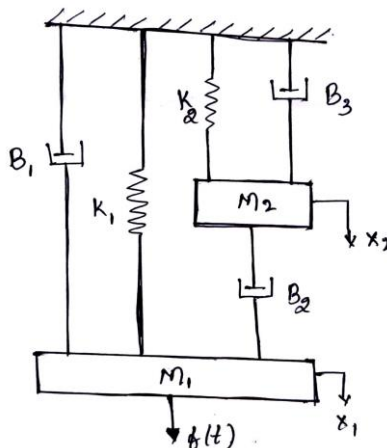


Fig.Q2(b)

(10 Marks)

Module-2

- 3 a. List and explain the various standard inputs used in control system analysis. (10 Marks)
- b. Develop an expression for steady state error for a simple closed loop control system. (10 Marks)

OR

- 4 a. Examine a 2nd order under damped system subjected to unit step input. (10 Marks)
- b. Evaluate the following quantities for a 2nd order unity feedback system with open loop

transfer function $G(s) = \frac{25}{s(s+7)}$, find :

- | | |
|---|--------------------|
| (i) Undamped natural frequency | (ii) Damping ratio |
| (iii) Damped natural frequency | (iv) Setting time |
| (v) Raise time | (v) Peak time |
| (vi) The percentage over shoot for unit step input. | |

(10 Marks)

Module-3

- 5 a. List the basic elements of block diagram. (05 Marks)
 b. Develop a closed loop transfer function for the block diagram shown in Fig.Q5(b).

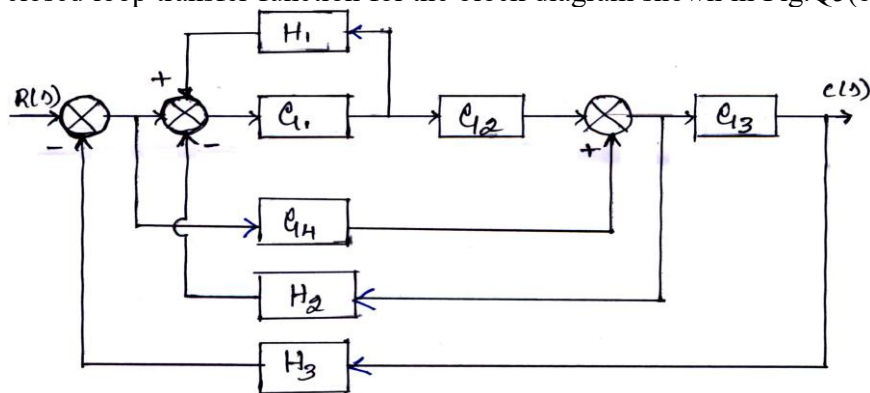


Fig.Q5(b)

(15 Marks)

OR

- 6 a. Explain the terms (i) Node (ii) Input node (iii) Output node (iv) Branch (v) Path. (05 Marks)
 b. Construct a S.F.G. for a block diagram shown in Fig.Q6(b) and obtain a transfer function using Mason's gain formula.

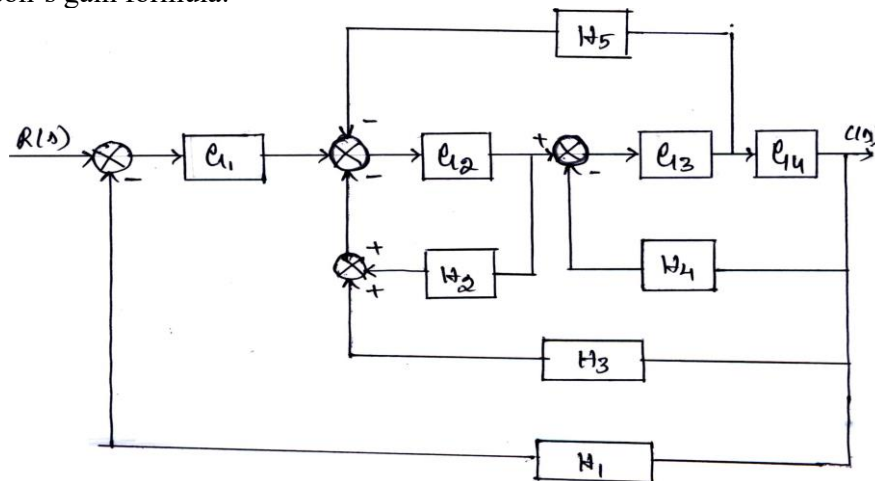


Fig.Q6(b)

(15 Marks)

Module-4

- 7 a. For a system with characteristic equation $F(s) = s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$. Examine stability using Routh Herwitz criterion. (10 Marks)
 b. A given system oscillates with frequency 2 rad/sec. Find values of K_{mar} and 'P' are in RHS.

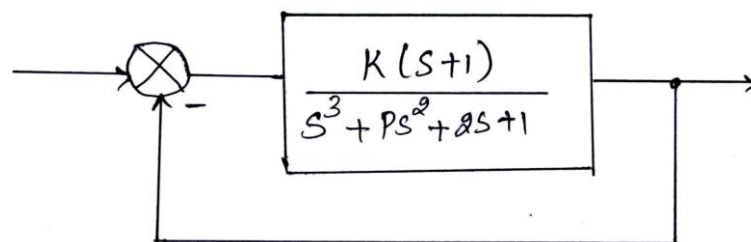


Fig.Q7(b)

(10 Marks)

OR

- 8** Construct a root locus for all value of 'K' ranging from 0 to ∞ for a feedback control system characterized by

$$G(s)H(s) = \frac{K}{s(s+4)(s^2+4s+20)} \quad (20 \text{ Marks})$$

Module-5

- 9** Using Nyquist criterion, examine the stability of a system whose open loop transfer function is

$$G(s)H(s) = \frac{K}{(s+1)(s+2)(s+3)} \quad (20 \text{ Marks})$$

OR

- 10** Construct a Bode plot for the following transfer function and determine gain margin and phase margin:

$$G(s)H(s) = \frac{10}{(1+s)(1+0.02s)} \quad (20 \text{ Marks})$$

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18ME72

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Computer Aided Design and Manufacturing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define CAD and CAM. Briefly explain the features of Fixed automation and programmable automation. (06 Marks)
- b. Illustrate the following with mathematical models:
 - (i) Production capacity
 - (ii) Work in process
 - (iii) TIP ratio(06 Marks)
- c. A batch manufacturing plant must be processed through 7 machines. There are 30 new batches. Average operation time is 8 min. Average setup time is 2 hours and non-operations time is 4 hours. Average batch size is 15 parts. Number of workstations is 10. The plant operates for an average of 125 hours/week. Determine manufacturing lead time, plant capacity and utilization. (08 Marks)

OR

- 2 a. What are automated flow lines? With sketches, explain Inline and Rotary type of automated flow lines. (08 Marks)
- b. Describe the methods of control of an automated flow line. (06 Marks)
- c. The following data applies to a 12 station inline transfer line, $p = 0.01$ for all the stations, cycle time is 0.3 min and repair time is 3 min using upper bound approach, compute the following:
 - (i) Frequency of line stops/cycle
 - (ii) Average production rate
 - (iii) Line efficiency(06 Marks)

Module-2

- 3 a. With block diagram, explain the various steps in computer aided design process. (08 Marks)
- b. Explain the functions of a graphics package. (06 Marks)
- c. Briefly explain Translation and Scaling. (06 Marks)

OR

- 4 a. Define Computer Aided Process Planning. List its benefits. (06 Marks)
- b. With block diagram, explain variant type of CAPP system. (08 Marks)
- c. What is shop floor control? Briefly explain the various phases of shop floor control. (06 Marks)

Module-3

- 5 a. With neat sketches, explain the types of FMS layouts. (10 Marks)
- b. Explain in brief with diagram the structure of Automated Storage and Retrieval System. (05 Marks)
- c. List the advantages of Group Technology. (05 Marks)

OR

- 6 a. Illustrate the following terms with reference to Line Balancing:

- (i) Minimum rational work element
- (ii) Precedence diagram
- (iii) Balance delay

(06 Marks)

- b. In a plant, a product is assembled as per the following data. Assume cycle time as 16 min:

Work element	1	2	3	4	5	6	7	8
T_e (min)	10	5	8	3	11	3	5	15
Preceded by	-	-	1, 2	2	3	3, 4	4	5, 6, 7

- (i) Construct precedence diagram
- (ii) Determine the number of stations required to balance the line by using LCR method.
- (iii) Determine balance delay.

(14 Marks)

Module-4

- 7 a. Briefly explain the classifications of CNC system.

(06 Marks)

- b. Write a note on cutter radius compensation.

(06 Marks)

- c. Write the part program for the part shape shown in Fig.Q7(c). Assume suitable machining parameters.

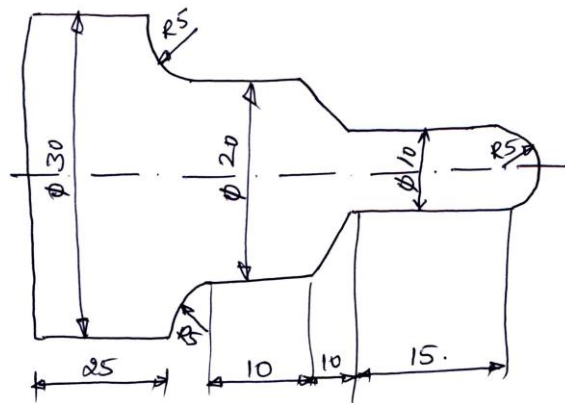


Fig.Q7(c) All dimensions are in mm

(08 Marks)

OR

- 8 a. With sketches, explain any three configurations of Industrial Robot.

(12 Marks)

- b. Write a note on various sensors used in Industrial Robot.

(08 Marks)

Module-5

- 9 a. With sketch, explain photopolymerization process.

(10 Marks)

- b. With sketch briefly explain Fused Deposition Modeling Technique.

(10 Marks)

OR

- 10 a. Briefly explain the various components of Industry 4.0.

(10 Marks)

- b. Write a note on Smart Manufacturing as applied to Industry 4.0.

(10 Marks)

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18ME734

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Total Quality Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Make the list of the basic concepts of TQM and briefly explain them. (10 Marks)
b. Enlighten on TQM frame work with a neat diagram. (10 Marks)

OR

- 2 a. Differentiate between American and Japan's approach towards quality. (10 Marks)
b. Elucidate on quality from customer's/consumer's and producer's perspective with a block diagram. (10 Marks)

Module-2

- 3 a. Which are the 12 characteristics of quality leaders? Describe them. (10 Marks)
b. Explain briefly the seven characteristics of effective people. (10 Marks)

OR

- 4 a. List out Deming's 14 points of TQM philosophy and explain any 4 of them. (10 Marks)
b. With Time-Management matrix, briefly describe put first things first. (10 Marks)

Module-3

- 5 a. What is Kano model? Why it is used? Explain the salient features. (10 Marks)
b. Who is customer? Define internal and external customer, with an example for each type. (10 Marks)

OR

- 6 a. How do you differentiate between dissatisfied, satisfied and delighted customers articulate with a sketch? (10 Marks)
b. Elucidate on customer perception of quality. (10 Marks)

Module-4

- 7 a. Sketch and explain Juran's Triology. (10 Marks)
b. Explicate on Scatter diagram and cause and effect diagram with a neat figures. (10 Marks)

OR

- 8 a. Explain PDSA cycle for continuous improvement with a neat figure. (10 Marks)
b. Enlighten on Histogram, Pareto diagram with a sketch and evidences. (10 Marks)

Module-5

- 9 a. Lean out steps involved in introduction of TPM. (10 Marks)
b. List and mention importance of 5's. (10 Marks)

OR

- 10 a. Which are the 8 pillars of TPM? Describe them with a neat sketch. (10 Marks)
b. Differentiate between productive maintenance and predictive maintenance with a table. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

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18ME744

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Mechatronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the scope and multi-disciplinary scenario of mechatronics with neat block diagrams. (10 Marks)
b. Explain with a neat sketch the working principle and characteristics of Linear Variable Differential Transformer (LVDT). (10 Marks)

OR

- 2 a. Explain with a neat diagram the elements of an engine management system. (10 Marks)
b. Explain with a neat diagram the working principle of Hall effect sensor. (10 Marks)

Module-2

- 3 a. Explain what is filtering and different types of filters with neat diagrams. (10 Marks)
b. Explain the following with neat sketches:
(i) Hybrid stepper motor
(ii) Pulse width modulation (10 Marks)

OR

- 4 a. Explain with a neat block diagram the objectives, functions and components (units) of a SCADA system. (10 Marks)
b. Explain with neat circuit diagrams the various types of DC motors with field coils and also their torque versus speed characteristics. (10 Marks)

Module-3

- 5 a. Explain with a neat block diagram the architecture of Intel's 8085A microprocessor. (15 Marks)
b. Distinguish between Microprocessor and Microcontroller. (05 Marks)

OR

- 6 Explain the following with neat diagrams:
a. Interrupt lines/pins
b. Input/output buffers
c. Program counter
d. Flags (20 Marks)

Module-4

- 7 a. Explain with a neat sketch the basic structure of PLC. (10 Marks)
b. Explain with neat sketches, the various symbols used in a ladder diagram. (10 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

OR

8 Explain the following pertaining to PLC:

- a. Shift registers
- b. Data handling
- c. Jump control
- d. Latching

(20 Marks)

Module-5

9 a. Explain with neat sketches the different types of guide ways.

(10 Marks)

b. Explain with a neat flow chart the stages of mechatronics design process.

(10 Marks)

OR

10 a. Explain the following with neat sketches:

- (i) Hydrostatic bearing
- (ii) Types of hydrodynamic bearings

(10 Marks)

b. Explain with a neat sketch the various parts or subsystems of pick and place robot. **(10 Marks)**

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18ME751

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Energy and Environment

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With relevant statistics enumerate the primary energy production trend in India. (10 Marks)
- b. List and explain various forms of energy in a brief manner. (05 Marks)
- c. Define energy management and explain energy demand estimation. (05 Marks)

OR

- 2 a. Elaborate the effect of various social and environmental aspects of India's energy development. (10 Marks)
- b. Define energy and power. Differentiate the same. (05 Marks)
- c. Interpret world energy scenario with respect to production and consumption using relevant statistics. (05 Marks)

Module-2

- 3 a. Define and explain sensible heat storage methods. (10 Marks)
- b. Explain in detail various phases of energy audit methodology. (10 Marks)

OR

- 4 a. Define and explain Latent heat storage methods. (10 Marks)
- b. Define energy audit and explain the need for energy audit. (05 Marks)
- c. Elaborate the benefits of thermal energy storage. (05 Marks)

Module-3

- 5 a. Define ecosystem and explain the following term :
 - i) Food chain
 - ii) Food web
 - iii) Ecological pyramid.(10 Marks)
- b. Enumerate how carbon cycle is utilized in the ecosystem. (05 Marks)
- c. Discuss how oxygen cycle is utilized in the ecosystems. (05 Marks)

OR

- 6 a. Define environment. Mention its scope. Discuss need for public awareness. (10 Marks)
- b. Elaborate how the nitrogen cycle ecosystem operates. (05 Marks)
- c. Discuss how water cycle is utilized in the ecosystem. (05 Marks)

Module-4

- 7 a. Discuss solid waste management techniques. (10 Marks)
b. Discuss briefly the causes effects and control measure of air pollution. (10 Marks)

OR

- 8 a. Enumerate the role of an individual in prevention of pollution. (10 Marks)
b. Enumerate the water pollution cases and its effects. Mention the control measures that can be initiated for mitigating the same. (10 Marks)

Module-5

- 9 a. Write a note on ozone layer depletion. (10 Marks)
b. Explain the role of Environment Impact Assessment (EIA) in reinforcing environmental legislation. (10 Marks)

OR

- 10 a. Discuss :
i) Wildlife Projection Act
ii) Forest Conservation Act. (10 Marks)
b. Enumerate the impact of global warming and acid rain on our mother nature. (10 Marks)

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